

Multiple Choice (10 marks)

1. What factors limit the number of trophic levels in a food chain?
 - (a) Constant energy levels across all trophic levels
 - (b) Increase in nutrient availability at each level
 - (c) Decrease in carbon dioxide concentration
 - (d) Increase in biomass at each successive level
 - (e) Great decrease (90%) in available energy at each level
2. An organism belonging to the nekton is which one of the following?
 - (a) Sponge
 - (b) Protist
 - (c) Clam
 - (d) Starfish
 - (e) Coral
3. Which of these is the smallest unit upon which natural selection directly acts?
 - (a) an individual's genome
 - (b) a population's genetic variation
 - (c) a species' genetic mutation
 - (d) an individual's genetic makeup
 - (e) a population's gene frequency
4. The first point of entry of water at the roots of a monocot plant is through the cytoplasm of cells of the
 - (a) parenchyma
 - (b) phloem
 - (c) endoderm
 - (d) pericycle
 - (e) cortex
5. What is an oncogene?
 - (a) Oncogenes are found only in viruses
 - (b) Oncogenes prevent cancer cells
 - (c) Oncogenes are a type of immune cell
 - (d) Oncogenes detoxify harmful substances in the cell
 - (e) Oncogenes slow down cell division

True / False (10 marks)

Answer True or False

6. True or False: Only inherited genetics and lifestyle factors can lead to cancer, excluding any contributions from viruses.
7. True or False: Rotational cleavage is characterized by early cell divisions occurring at an oblique angle to the polar axis.
8. True or False: Sound communication in animals is primarily used for mineral absorption and heat regulation.
9. True or False: The probability of drawing the ace of diamonds from a standard deck of cards is $1/53$.
10. True or False: The embryos of terrestrial vertebrates are encased within a fluid secreted by the chorion to facilitate nutrient absorption.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the adaptive features of animals living in hot arid environments, focusing on the role of the loop of Henle in water conservation and its evolutionary significance.
12. Discuss the mechanisms that ensure the proper routing of food and air through the pharynx, highlighting the roles of the epiglottis and the swallowing reflex.

End of Paper